

# Cloud, Programming & Generative AI

## MCQ Bank

250 Questions with Answers

*Answers shown in muted gray — grouped by topic*

### Difficulty Legend:

● Easy ● Medium ● Difficult

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# Cloud Fundamentals

20 questions

## Q1. What does 'cloud computing' primarily refer to?

[Easy]

- A. A weather-forecasting software package
- B. On-demand delivery of computing resources (servers, storage, databases) over the internet
- C. A type of physical network cable
- D. Storing files only on a local hard drive

Ans: B. On-demand delivery of computing resources (servers, storage, databases) over the internet

## Q2. Which cloud service model provides ready-to-use software applications over the internet?

[Easy]

- A. DaaS (Data as a Service)
- B. SaaS (Software as a Service)
- C. IaaS (Infrastructure as a Service)
- D. PaaS (Platform as a Service)

Ans: B. SaaS (Software as a Service)

## Q3. In a public cloud deployment model, the infrastructure is:

[Easy]

- A. Dedicated to a single organization and hosted on-premises only
- B. Limited to storing printed documents
- C. Never connected to the internet
- D. Owned and operated by a third-party provider and shared across multiple customers

Ans: D. Owned and operated by a third-party provider and shared across multiple customers

## Q4. Which characteristic best describes 'elasticity' in cloud computing?

[Medium]

- A. Resources automatically scale up or down to match demand
- B. All servers must be manually restarted daily
- C. Data can never be deleted once uploaded
- D. Applications can only run in one geographic region

Ans: A. Resources automatically scale up or down to match demand

## Q5. What is a key difference between IaaS and PaaS?

[Medium]

- A. PaaS gives you raw hardware while IaaS gives finished apps
- B. They are identical with different names

- C. IaaS can only run on-premises and PaaS only in the cloud
- D. IaaS provides virtualized infrastructure (VMs, storage, networking) while PaaS also provides a managed platform/runtime for building and deploying apps

Ans: D. IaaS provides virtualized infrastructure (VMs, storage, networking) while PaaS also provides a managed platform/runtime for building and deploying apps

**Q6. In the shared responsibility model, who is typically responsible for securing customer data and access management in a cloud service?**

[Medium]

- A. The customer, while the provider secures the underlying infrastructure
- B. An external auditor who owns all the data
- C. Neither party is responsible for data security
- D. The cloud provider alone

Ans: A. The customer, while the provider secures the underlying infrastructure

**Q7. An organization needs strict data residency and dedicated control for regulated workloads, but also wants burst capacity for seasonal peaks. Which deployment approach best fits?**

[Difficult]

- A. Run all workloads on individual employee laptops
- B. Adopt a hybrid cloud: keep regulated workloads in a private cloud/on-prem while bursting non-sensitive peak load to public cloud, connected securely
- C. Move everything to a single public cloud region and ignore residency rules
- D. Use only a community cloud with no private component

Ans: B. Adopt a hybrid cloud: keep regulated workloads in a private cloud/on-prem while bursting non-sensitive peak load to public cloud, connected securely

**Q8. A team wants to run event-driven code without managing servers, paying only for actual execution time. Which cloud model is most appropriate and what is its trade-off?**

[Difficult]

- A. Serverless / Function-as-a-Service, with trade-offs such as cold-start latency and execution time limits
- B. Static file hosting, which can execute arbitrary backend logic
- C. Dedicated bare-metal servers, with the trade-off of instant scaling
- D. A traditional on-premises data center, with no operational overhead

Ans: A. Serverless / Function-as-a-Service, with trade-offs such as cold-start latency and execution time limits

**Q81. What is a defining benefit of cloud computing's pay-as-you-go pricing?**

[Easy]

- A. Only free resources are available
- B. You must pre-purchase 10 years of capacity
- C. Pricing is fixed regardless of usage

D. You pay only for the resources you actually consume

Ans: D. You pay only for the resources you actually consume

**Q82. What is a 'region' in a cloud provider's infrastructure?**

[Easy]

- A. A type of virtual machine
- B. A single physical server
- C. A billing invoice
- D. A geographic area containing one or more data centers

Ans: D. A geographic area containing one or more data centers

**Q83. Which of these is an example of Infrastructure as a Service (IaaS)?**

[Easy]

- A. A managed no-code app builder
- B. A ready-to-use email application
- C. Renting virtual machines and storage that you configure yourself
- D. A finished CRM website

Ans: C. Renting virtual machines and storage that you configure yourself

**Q84. What does 'scalability' mean for a cloud application?**

[Easy]

- A. A fixed number of users forever
- B. Automatic deletion of data
- C. The inability to grow beyond one server
- D. The ability to handle increased load by adding resources

Ans: D. The ability to handle increased load by adding resources

**Q85. What is the difference between vertical scaling and horizontal scaling?**

[Medium]

- A. Vertical scaling adds more power (CPU/RAM) to an existing machine; horizontal scaling adds more machines
- B. Horizontal scaling always reduces cost to zero
- C. Vertical scaling only works on databases
- D. They are the same thing

Ans: A. Vertical scaling adds more power (CPU/RAM) to an existing machine; horizontal scaling adds more machines

**Q86. What is a private cloud?**

[Medium]

- A. A social network
- B. A public website open to everyone
- C. A type of encryption algorithm

D. Cloud infrastructure dedicated to and controlled by a single organization

Ans: D. Cloud infrastructure dedicated to and controlled by a single organization

**Q87. What is the primary purpose of a Content Delivery Network (CDN)?**

[Medium]

- A. To cache and serve content from edge locations close to users, reducing latency
- B. To compile source code
- C. To store passwords
- D. To replace the database

Ans: A. To cache and serve content from edge locations close to users, reducing latency

**Q88. In cloud computing, what does 'high availability' aim to achieve?**

[Medium]

- A. Guaranteeing zero cost
- B. Storing all data on one disk
- C. Preventing any users from connecting
- D. Minimizing downtime so services remain accessible despite failures

Ans: D. Minimizing downtime so services remain accessible despite failures

**Q89. What is 'multi-tenancy' in a SaaS platform?**

[Medium]

- A. A single software instance serves multiple customers while isolating their data logically
- B. Tenants must share their data publicly
- C. Only one customer can use the app at a time
- D. Each customer gets a separate physical building

Ans: A. A single software instance serves multiple customers while isolating their data logically

**Q90. A company must design for disaster recovery with a very low Recovery Time Objective (RTO). Which strategy best fits?**

[Difficult]

- A. Take a monthly backup and restore manually when needed
- B. Rely on employee laptops as the only backup
- C. Maintain a warm or hot standby in a second region with automated failover so services resume quickly
- D. Disable backups to save storage

Ans: C. Maintain a warm or hot standby in a second region with automated failover so services resume quickly

**Q91. Why might an organization choose object storage over block storage for a data lake?**

[Difficult]

- A. Object storage scales massively, is cost-effective for unstructured data, and is accessed via APIs—ideal for large analytics datasets
- B. Object storage guarantees SQL transactions

- C. Block storage cannot store any files
- D. Object storage is limited to a few megabytes

Ans: A. Object storage scales massively, is cost-effective for unstructured data, and is accessed via APIs—ideal for large analytics datasets

**Q92. A microservices app suffers cascading failures when one downstream service slows down. Which cloud-native resilience pattern helps most?**

**[Difficult]**

- A. Scale down to a single instance
- B. Route all traffic to the failing service
- C. Remove all monitoring
- D. Apply circuit breakers, timeouts, and bulkheads to isolate failures and prevent one slow service from exhausting resources everywhere

Ans: D. Apply circuit breakers, timeouts, and bulkheads to isolate failures and prevent one slow service from exhausting resources everywhere

# RDBMS & SQL (MySQL)

35 questions

**Q9. What does RDBMS stand for?**

[Easy]

- A. Rapid Data Backup Management Service
- B. Remote Distributed Batch Messaging System
- C. Relational Database Management System
- D. Reliable Directory of Business Metrics Store

Ans: C. Relational Database Management System

**Q10. In a relational database, what is a 'primary key'?**

[Easy]

- A. The first column alphabetically in any table
- B. A key used only to encrypt data
- C. A column (or set of columns) that uniquely identifies each row in a table
- D. The password used to log into MySQL

Ans: C. A column (or set of columns) that uniquely identifies each row in a table

**Q11. Which SQL statement is used to retrieve data from a table?**

[Easy]

- A. UPDATE
- B. SELECT
- C. DELETE
- D. INSERT

Ans: B. SELECT

**Q12. Which SQL clause is used to filter rows based on a condition?**

[Easy]

- A. HAVING
- B. WHERE
- C. GROUP BY
- D. ORDER BY

Ans: B. WHERE

**Q13. What does a FOREIGN KEY enforce in a relational schema?**

[Medium]

- A. Automatic encryption of the referenced column
- B. Referential integrity by linking a column to the primary key of another table
- C. It guarantees a column contains only numbers

D. That a table can never be deleted

Ans: B. Referential integrity by linking a column to the primary key of another table

**Q14. Which JOIN returns only the rows that have matching values in both tables?**

[Medium]

- A. INNER JOIN
- B. FULL OUTER JOIN
- C. RIGHT JOIN
- D. LEFT JOIN

Ans: A. INNER JOIN

**Q15. What is the difference between WHERE and HAVING in SQL?**

[Medium]

- A. They are interchangeable in all cases
- B. WHERE filters rows before aggregation; HAVING filters groups after aggregation (e.g., on aggregate results)
- C. WHERE only works on numeric columns
- D. HAVING can only be used without GROUP BY

Ans: B. WHERE filters rows before aggregation; HAVING filters groups after aggregation (e.g., on aggregate results)

**Q16. Which aggregate function would you use to count the number of rows in a result set?**

[Medium]

- A. MAX()
- B. SUM()
- C. AVG()
- D. COUNT()

Ans: D. COUNT()

**Q17. What is database normalization primarily intended to reduce?**

[Medium]

- A. Data redundancy and update anomalies
- B. The need for primary keys
- C. The number of tables to exactly one
- D. Query readability

Ans: A. Data redundancy and update anomalies

**Q18. In MySQL, what does the GROUP BY clause do?**

[Medium]

- A. Groups rows that share values in specified columns so aggregate functions can be applied per group



- B. Creates a foreign key automatically
- C. Sorts the final result in ascending order
- D. Deletes duplicate tables

Ans: A. Groups rows that share values in specified columns so aggregate functions can be applied per group

**Q19. A query joins a large orders table to a customers table and runs slowly. Filtering is done on orders.customer\_id. What is the most effective first optimization?**

[Difficult]

- A. Convert both tables to a single denormalized text file
- B. Remove the WHERE clause entirely
- C. Create an index on the join/filter column (e.g., orders.customer\_id) so the engine can avoid full table scans
- D. Add SELECT \* to return more columns

Ans: C. Create an index on the join/filter column (e.g., orders.customer\_id) so the engine can avoid full table scans

**Q20. You need each department's highest-paid employee along with their name. Which approach correctly returns this?**

[Difficult]

- A. SELECT dept, MAX(salary), name GROUP BY dept — because non-aggregated columns are always valid
- B. Use a window function (e.g., ROW\_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)) or a correlated subquery, since selecting name alongside a bare MAX() per group is ambiguous
- C. A simple SELECT \* with no grouping
- D. DELETE the lower-paid employees first

Ans: B. Use a window function (e.g., ROW\_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)) or a correlated subquery, since selecting name alongside a bare MAX() per group is ambiguous

**Q21. A LEFT JOIN between customers and orders is being used to find customers with NO orders, but adding a WHERE orders.amount > 100 filter unexpectedly drops the no-order customers. Why?**

[Difficult]

- A. LEFT JOIN never returns unmatched rows
- B. MySQL does not support LEFT JOIN
- C. The customers table must be indexed for LEFT JOIN to work at all
- D. A condition on the right table's columns in WHERE turns the outer join into an effective inner join; the null-row customers fail the predicate. The 'no orders' check should use WHERE orders.id IS NULL, and amount filters belong in the ON clause

Ans: D. A condition on the right table's columns in WHERE turns the outer join into an effective inner join; the null-row customers fail the predicate. The 'no orders' check should use WHERE orders.id IS NULL, and amount filters belong in the ON clause

**Q22. Two concurrent transactions each read a balance, then update it, causing one update to be lost. Which concept and mechanism address this?**

[Difficult]

- A. This is a syntax error unrelated to concurrency
- B. Drop the primary key to allow simultaneous writes
- C. Simply run both queries faster to avoid overlap
- D. It is a lost-update / concurrency problem; use transactions with appropriate isolation levels and locking (or optimistic concurrency with version checks) to enforce consistency

Ans: D. It is a lost-update / concurrency problem; use transactions with appropriate isolation levels and locking (or optimistic concurrency with version checks) to enforce consistency

**Q93. Which SQL statement adds new rows to a table?**

[Easy]

- A. ALTER
- B. SELECT
- C. DROP
- D. INSERT

Ans: D. INSERT

**Q94. Which SQL statement removes rows from a table?**

[Easy]

- A. SELECT
- B. CREATE
- C. INDEX
- D. DELETE

Ans: D. DELETE

**Q95. What does the DISTINCT keyword do in a SELECT query?**

[Easy]

- A. Joins two tables
- B. Returns only unique (non-duplicate) values
- C. Deletes duplicate tables
- D. Sorts rows alphabetically

Ans: B. Returns only unique (non-duplicate) values

**Q96. Which clause sorts the result set of a query?**

[Easy]

- A. HAVING
- B. GROUP BY
- C. ORDER BY
- D. WHERE

Ans: C. ORDER BY

**Q97. In MySQL, which data type is best suited for storing whole numbers?**

[Easy]

- A. VARCHAR
- B. INT
- C. BLOB
- D. DATE

Ans: B. INT

**Q98. What does a 'row' (record) represent in a relational table?**

[Easy]

- A. The name of the database
- B. A single column definition
- C. A single entity/instance with values for each column
- D. An index file

Ans: C. A single entity/instance with values for each column

**Q99. What is the purpose of a UNIQUE constraint?**

[Medium]

- A. To ensure all values in a column (or set) are distinct
- B. To allow unlimited duplicate values
- C. To encrypt the column
- D. To delete null values

Ans: A. To ensure all values in a column (or set) are distinct

**Q100. What does a LEFT JOIN return?**

[Medium]

- A. All rows from the left table plus matching rows from the right (nulls where no match)
- B. Only matching rows in both tables
- C. No rows at all
- D. Only rows from the right table

Ans: A. All rows from the left table plus matching rows from the right (nulls where no match)

**Q101. What is a subquery?**

[Medium]

- A. A user permission
- B. A backup of the database
- C. A query nested inside another query
- D. A type of index

Ans: C. A query nested inside another query

**Q102. What does the ACID acronym describe in databases?**

[Medium]

- A. A programming language
- B. A type of storage disk
- C. A query optimization tool
- D. Atomicity, Consistency, Isolation, Durability—properties guaranteeing reliable transactions

Ans: D. Atomicity, Consistency, Isolation, Durability—properties guaranteeing reliable transactions

**Q103. What is the difference between DELETE and TRUNCATE?**

[Medium]

- A. DELETE drops the whole table
- B. They are identical
- C. DELETE removes rows (optionally filtered) and is logged/rollback-able; TRUNCATE quickly removes all rows and resets the table with minimal logging
- D. TRUNCATE only deletes one row

Ans: C. DELETE removes rows (optionally filtered) and is logged/rollback-able; TRUNCATE quickly removes all rows and resets the table with minimal logging

**Q104. What does an index primarily improve?**

[Medium]

- A. The speed of data retrieval on indexed columns
- B. The size of the disk to zero
- C. The number of tables
- D. The color of the output

Ans: A. The speed of data retrieval on indexed columns

**Q105. What is a composite key?**

[Medium]

- A. A key that is always encrypted
- B. A key made of two or more columns that together uniquely identify a row
- C. A duplicate primary key
- D. A foreign key to itself

Ans: B. A key made of two or more columns that together uniquely identify a row

**Q106. What does the SQL keyword LIKE with a wildcard '%' do?**

[Medium]

- A. Deletes matching rows
- B. Creates an index
- C. Matches a pattern in text (e.g., 'A%' matches strings starting with A)
- D. Performs an exact numeric match

Ans: C. Matches a pattern in text (e.g., 'A%' matches strings starting with A)

**Q107. What is a VIEW in SQL?**

[Medium]

- A. A user login screen
- B. A physical copy of a table on disk
- C. A stored, named query that presents data like a virtual table
- D. A backup file

Ans: C. A stored, named query that presents data like a virtual table

**Q108. A report needs a running total of sales ordered by date. Which SQL feature is most appropriate?**

[Difficult]

- A. A simple GROUP BY with no ordering
- B. A window function such as SUM(sales) OVER (ORDER BY date)
- C. A foreign key constraint
- D. Deleting older rows

Ans: B. A window function such as SUM(sales) OVER (ORDER BY date)

**Q109. Third Normal Form (3NF) requires eliminating which dependency?**

[Difficult]

- A. Transitive dependencies—non-key attributes must depend only on the primary key, not on other non-key attributes
- B. All relationships between tables
- C. Every primary key
- D. All numeric columns

Ans: A. Transitive dependencies—non-key attributes must depend only on the primary key, not on other non-key attributes

**Q110. A query uses a function on an indexed column in the WHERE clause (e.g., WHERE YEAR(order\_date)=2025) and ignores the index. Why, and what is the fix?**

[Difficult]

- A. MySQL does not support indexes on dates
- B. The fix is to remove the WHERE clause
- C. Indexes never work with dates
- D. Wrapping the column in a function makes it non-sargable, so the index can't be used; rewrite as a range (order\_date >= '2025-01-01' AND < '2026-01-01') to keep it sargable

Ans: D. Wrapping the column in a function makes it non-sargable, so the index can't be used; rewrite as a range (order\_date >= '2025-01-01' AND < '2026-01-01') to keep it sargable

**Q111. What is a deadlock in a relational database and a common mitigation?**

[Difficult]

- A. A syntax error in SELECT
- B. A missing primary key

- C. A permanent hardware failure
- D. Two transactions each hold a lock the other needs, blocking both; mitigate by accessing resources in a consistent order, keeping transactions short, and handling deadlock retries

Ans: D. Two transactions each hold a lock the other needs, blocking both; mitigate by accessing resources in a consistent order, keeping transactions short, and handling deadlock retries

**Q112. When would a covering index significantly speed up a query?**

[Difficult]

- A. Only for INSERT statements
- B. When the index contains all columns the query needs, so the engine answers it from the index without reading the table rows
- C. Indexes never help SELECT queries
- D. When the query returns SELECT \*

Ans: B. When the index contains all columns the query needs, so the engine answers it from the index without reading the table rows

**Q113. A star-schema data warehouse separates data into fact and dimension tables. What is the main benefit for analytics?**

[Difficult]

- A. It removes the need for indexes
- B. It organizes measurable events (facts) around descriptive context (dimensions), enabling efficient, intuitive analytical queries and aggregations
- C. It stores only one row
- D. It prevents all joins

Ans: B. It organizes measurable events (facts) around descriptive context (dimensions), enabling efficient, intuitive analytical queries and aggregations

# Python for Data Engineers

40 questions

**Q23. Which Python data structure is ordered and mutable (can be changed after creation)?**

[Easy]

- A. list
- B. int
- C. frozenset
- D. tuple

Ans: A. list

**Q24. Which library is most commonly used for tabular data analysis in Python?**

[Easy]

- A. turtle
- B. tkinter
- C. pandas
- D. socket

Ans: C. pandas

**Q25. What is a Python dictionary?**

[Easy]

- A. A function that returns text
- B. An ordered list of numbers only
- C. A collection of key-value pairs
- D. A type of loop

Ans: C. A collection of key-value pairs

**Q26. Which keyword is used to define a function in Python?**

[Easy]

- A. def
- B. define
- C. func
- D. function

Ans: A. def

**Q27. In pandas, what does `df.groupby('col').agg({'x':'sum'})` do?**

[Medium]

- A. Renames the DataFrame to 'col'
- B. Sorts the DataFrame by 'x'

- C. Groups rows by 'col' and computes the sum of column 'x' for each group
- D. Deletes the column 'col'

Ans: C. Groups rows by 'col' and computes the sum of column 'x' for each group

**Q28. What is the primary advantage of NumPy arrays over standard Python lists for numerical work?**

[Medium]

- A. They offer vectorized operations and contiguous, typed memory for faster numerical computation
- B. They cannot be indexed
- C. They can only hold strings
- D. They automatically connect to databases

Ans: A. They offer vectorized operations and contiguous, typed memory for faster numerical computation

**Q29. Which pandas method would you use to combine two DataFrames based on a common key column, similar to a SQL join?**

[Medium]

- A. df.dropna()
- B. pd.merge()
- C. df.plot()
- D. df.describe()

Ans: B. pd.merge()

**Q30. When reading a large CSV that doesn't fit comfortably in memory, which pandas approach helps?**

[Medium]

- A. Read it twice to be safe
- B. Use the chunksize parameter (or specify dtypes / usecols) to process the file in smaller pieces
- C. Convert it to an image first
- D. Disable all data types

Ans: B. Use the chunksize parameter (or specify dtypes / usecols) to process the file in smaller pieces

**Q31. What does the context manager 'with open(file) as f:' primarily ensure?**

[Medium]

- A. The file is encrypted
- B. The file is deleted after reading
- C. The file is uploaded to the cloud
- D. The file is automatically and properly closed when the block ends, even if an error occurs

Ans: D. The file is automatically and properly closed when the block ends, even if an error occurs

**Q32. In pandas, how are missing values typically represented and one common way to handle them?**



[Medium]

- A. As the string 'missing'; they cannot be handled
- B. As boolean True; handled by renaming columns
- C. As 0 always; they must be ignored
- D. As NaN; handled via methods like `dropna()` to remove or `fillna()` to impute

Ans: D. As NaN; handled via methods like `dropna()` to remove or `fillna()` to impute

**Q33. A data pipeline uses `df.apply(func, axis=1)` row-by-row and is far too slow on millions of rows. What is the most effective Pythonic fix?**

[Difficult]

- A. Add more print statements to monitor progress
- B. Replace the row-wise apply with vectorized pandas/NumPy operations (or built-in methods), which operate on whole arrays in C rather than per-row Python calls
- C. Convert the DataFrame to a list of dictionaries and loop manually
- D. Increase the loop counter

Ans: B. Replace the row-wise apply with vectorized pandas/NumPy operations (or built-in methods), which operate on whole arrays in C rather than per-row Python calls

**Q34. Why can `'a = [[0]*3]*3'` create a bug when you set `a[0][0] = 1` and multiple rows change?**

[Difficult]

- A. The outer multiplication copies references to the same inner list, so all rows point to one list; use a list comprehension like `[[0]*3 for _ in range(3)]` to create independent rows
- B. Because the syntax raises a `SyntaxError`
- C. Because lists cannot hold integers
- D. Because Python lists are immutable

Ans: A. The outer multiplication copies references to the same inner list, so all rows point to one list; use a list comprehension like `[[0]*3 for _ in range(3)]` to create independent rows

**Q35. A generator function using 'yield' is chosen over building a full list. What is the key engineering benefit for large data streams?**

[Difficult]

- A. It produces items lazily one at a time, keeping memory usage low and enabling streaming over datasets larger than RAM
- B. It automatically parallelizes across all CPU cores
- C. It loads all data into memory at once for speed
- D. It converts the data to SQL

Ans: A. It produces items lazily one at a time, keeping memory usage low and enabling streaming over datasets larger than RAM

**Q36. You must ensure a numeric ID column loaded from CSV keeps leading zeros and isn't inferred as int. What is the correct pandas practice?**

[Difficult]

- A. Specify the column dtype as str on read (e.g., dtype={'id': str}) so pandas doesn't infer it as an integer and strip the zeros
- B. Multiply the column by 10 after loading
- C. Convert the whole DataFrame to floats
- D. Nothing; pandas always preserves leading zeros

Ans: A. Specify the column dtype as str on read (e.g., dtype={'id': str}) so pandas doesn't infer it as an integer and strip the zeros

**Q114. Which Python data structure stores unique, unordered items?**

[Easy]

- A. str
- B. tuple
- C. list
- D. set

Ans: D. set

**Q115. Which symbol starts a single-line comment in Python?**

[Easy]

- A. //
- B. --
- C. /\*
- D. #

Ans: D. #

**Q116. What does the len() function return?**

[Easy]

- A. The largest value
- B. The number of items in an object (e.g., list length)
- C. A random number
- D. The sum of items

Ans: B. The number of items in an object (e.g., list length)

**Q117. How do you create a tuple in Python?**

[Easy]

- A. Using square brackets [ ]
- B. Using parentheses ( ) with comma-separated values
- C. Using angle brackets < >
- D. Using curly braces { }

Ans: B. Using parentheses ( ) with comma-separated values

**Q118. Which loop repeats a block while a condition is true?**

[Easy]

- A. switch
- B. for-only
- C. goto
- D. while

Ans: D. while

**Q119. What does the import statement do?**

[Easy]

- A. Loads a module/library so its functions can be used
- B. Prints text
- C. Deletes a module
- D. Defines a class

Ans: A. Loads a module/library so its functions can be used

**Q120. In pandas, what is a DataFrame?**

[Easy]

- A. A network socket
- B. A plotting window
- C. A single number
- D. A 2D labeled tabular data structure with rows and columns

Ans: D. A 2D labeled tabular data structure with rows and columns

**Q121. Which method reads a CSV file into a pandas DataFrame?**

[Easy]

- A. `pd.open_file()`
- B. `pd.read_csv()`
- C. `pd.get_csv()`
- D. `pd.load_table()`

Ans: B. `pd.read_csv()`

**Q122. What does list comprehension provide in Python?**

[Medium]

- A. A type of exception
- B. A way to connect to databases
- C. A method to delete files
- D. A concise way to build a list from an iterable with optional filtering

Ans: D. A concise way to build a list from an iterable with optional filtering

**Q123. What is the difference between a list and a tuple?**

[Medium]

- A. Lists are mutable; tuples are immutable

- B. Tuples can grow but lists cannot
- C. Lists can only hold strings
- D. They are exactly the same

Ans: A. Lists are mutable; tuples are immutable

**Q124. In pandas, what does `df.loc[]` vs `df.iloc[]` differ by?**

[Medium]

- A. `loc` deletes rows
- B. `loc` selects by label/boolean; `iloc` selects by integer position
- C. They are identical
- D. `iloc` only works on strings

Ans: B. `loc` selects by label/boolean; `iloc` selects by integer position

**Q125. What does a Python virtual environment (venv) provide?**

[Medium]

- A. An isolated set of installed packages for a project, avoiding dependency conflicts
- B. A cloud database
- C. A GUI framework
- D. A faster CPU

Ans: A. An isolated set of installed packages for a project, avoiding dependency conflicts

**Q126. What does the `try/except` block handle?**

[Medium]

- A. Faster loops
- B. Exceptions/errors, allowing graceful handling instead of crashing
- C. Variable naming
- D. Importing modules

Ans: B. Exceptions/errors, allowing graceful handling instead of crashing

**Q127. In pandas, what does `df.merge(other, how='left', on='id')` produce?**

[Medium]

- A. A pivot table
- B. A left join keeping all rows of `df` and matching rows of `other` on 'id'
- C. An inner join only
- D. A deletion of `df`

Ans: B. A left join keeping all rows of `df` and matching rows of `other` on 'id'

**Q128. What is the purpose of the pandas method `df.pivot_table()`?**

[Medium]

- A. To compile code
- B. To summarize/aggregate data across dimensions, similar to a spreadsheet pivot

- C. To open a socket
- D. To rename the file

Ans: B. To summarize/aggregate data across dimensions, similar to a spreadsheet pivot

**Q129. What is the difference between '==' and 'is' in Python?**

[Medium]

- A. '==' checks memory address
- B. They are identical
- C. '==' compares values; 'is' checks whether two references point to the same object
- D. 'is' compares values only

Ans: C. '==' compares values; 'is' checks whether two references point to the same object

**Q130. What is a lambda function?**

[Medium]

- A. A database cursor
- B. A small anonymous function defined inline
- C. A file handler
- D. A class definition

Ans: B. A small anonymous function defined inline

**Q131. Which format is commonly used for columnar, compressed storage in data engineering?**

[Medium]

- A. BMP
- B. Parquet
- C. MP3
- D. DOCX

Ans: B. Parquet

**Q132. Chained indexing like `df[df.a>0]['b']=1` raises a `SettingWithCopyWarning`. Why, and what is the correct approach?**

[Difficult]

- A. The warning means the file is corrupt
- B. Chained indexing always errors and stops execution
- C. There is no correct approach; pandas cannot assign values
- D. It may operate on a temporary copy so the assignment might not stick; use `df.loc[df.a>0,'b']=1` for a single, well-defined assignment

Ans: D. It may operate on a temporary copy so the assignment might not stick; use `df.loc[df.a>0,'b']=1` for a single, well-defined assignment

**Q133. A default mutable argument like `def f(x, acc=[])` accumulates values across calls unexpectedly. Why?**

[Difficult]

- A. It is a bug only in old Python 2
- B. Default arguments are evaluated once at function definition, so the same list is reused; use `acc=None` and create a new list inside the function
- C. Python creates a new list every call automatically
- D. Lists cannot be default arguments

Ans: B. Default arguments are evaluated once at function definition, so the same list is reused; use `acc=None` and create a new list inside the function

**Q134. Why can floating-point sums like `0.1+0.2` not equal exactly `0.3`, and how should equality be handled?**

[Difficult]

- A. Binary floating point cannot represent some decimals exactly; compare with a tolerance (e.g., `math.isclose`) or use the decimal module for exact arithmetic
- B. Floats always equal their decimal literal
- C. Because integers overflow
- D. Because Python is broken

Ans: A. Binary floating point cannot represent some decimals exactly; compare with a tolerance (e.g., `math.isclose`) or use the decimal module for exact arithmetic

**Q135. For a CPU-bound data transformation, why might Python threads not speed it up, and what helps?**

[Difficult]

- A. Only I/O-bound tasks are slowed by threads
- B. The GIL serializes CPU-bound bytecode across threads; use multiprocessing (or vectorized/native libraries) to leverage multiple cores
- C. Threads always double CPU speed
- D. Python has no threads

Ans: B. The GIL serializes CPU-bound bytecode across threads; use multiprocessing (or vectorized/native libraries) to leverage multiple cores

**Q136. When processing a 100 GB file that won't fit in memory, which strategy is soundest?**

[Difficult]

- A. Load it all with `pd.read_csv()` at once
- B. Stream/iterate in chunks (or use out-of-core tools like Dask/PySpark, or columnar formats), processing incrementally
- C. Rename the file to reduce its size
- D. Convert it to a Python list first

Ans: B. Stream/iterate in chunks (or use out-of-core tools like Dask/PySpark, or columnar formats), processing incrementally

**Q137. Why is using vectorized string operations (`df['s'].str.lower()`) preferred over a Python loop in pandas?**

[Difficult]

- A. Vectorization only works on integers
- B. They run optimized C-level operations across the whole Series, avoiding slow per-element Python overhead
- C. str methods encrypt the data
- D. Loops are impossible in Python

Ans: B. They run optimized C-level operations across the whole Series, avoiding slow per-element Python overhead

**Q138. In an ETL job, why is idempotency an important design property?**

[Difficult]

- A. It ensures rerunning the same job produces the same result without duplicating or corrupting data, enabling safe retries
- B. It makes the job run only once ever
- C. It doubles the output on each run
- D. It deletes the source data

Ans: A. It ensures rerunning the same job produces the same result without duplicating or corrupting data, enabling safe retries

**Q139. What problem does schema validation (e.g., with Pydantic or a schema check) solve in a data pipeline?**

[Difficult]

- A. It catches malformed or unexpected data early, enforcing types and structure before it corrupts downstream systems
- B. It speeds up the CPU
- C. It replaces the database
- D. It compresses files

Ans: A. It catches malformed or unexpected data early, enforcing types and structure before it corrupts downstream systems

# App Interfaces & APIs

13 questions

## Q78. What is Streamlit primarily used for?

[Medium]

- A. Rapidly building interactive data/ML web apps and dashboards in Python with minimal front-end code
- B. Compiling low-level C programs
- C. Encrypting network traffic
- D. Managing relational database transactions

Ans: A. Rapidly building interactive data/ML web apps and dashboards in Python with minimal front-end code

## Q79. A team exposes a Python ML model to other services and needs a high-performance, well-documented HTTP API with request validation. Why is FastAPI a strong fit?

[Difficult]

- A. Because it is a database engine
- B. Because it only serves static HTML
- C. FastAPI provides async support, automatic data validation via type hints/Pydantic, and auto-generated interactive API docs (OpenAPI/Swagger), making robust APIs quick to build
- D. Because it prevents any API documentation

Ans: C. FastAPI provides async support, automatic data validation via type hints/Pydantic, and auto-generated interactive API docs (OpenAPI/Swagger), making robust APIs quick to build

## Q80. When would you choose FastAPI over Streamlit for delivering a GenAI feature?

[Difficult]

- A. They are identical and interchangeable in every scenario
- B. Always use Streamlit because APIs are unnecessary
- C. FastAPI is only for building spreadsheets
- D. Use FastAPI when you need a programmatic backend/API for other apps or scalable services, and Streamlit when you need a quick interactive UI for humans; they solve different layers and are often combined

Ans: D. Use FastAPI when you need a programmatic backend/API for other apps or scalable services, and Streamlit when you need a quick interactive UI for humans; they solve different layers and are often combined

## Q241. What is an API?

[Easy]

- A. A monitor
- B. A physical cable
- C. A spreadsheet
- D. A defined interface that lets software components communicate and exchange data

Ans: D. A defined interface that lets software components communicate and exchange data



**Q242. What does Streamlit let you build with mostly Python?**

[Easy]

- A. Printer firmware
- B. Interactive web apps and dashboards
- C. Network drivers
- D. Operating systems

Ans: B. Interactive web apps and dashboards

**Q243. What is JSON commonly used for in APIs?**

[Easy]

- A. Encrypting hard drives
- B. Rendering 3D games
- C. Compiling C code
- D. A lightweight, human-readable format for exchanging structured data

Ans: D. A lightweight, human-readable format for exchanging structured data

**Q244. What HTTP method is typically used to create a new resource via a REST API?**

[Medium]

- A. GET only
- B. DELETE
- C. No method needed
- D. POST

Ans: D. POST

**Q245. What does FastAPI generate automatically from your type-annotated endpoints?**

[Medium]

- A. Interactive OpenAPI/Swagger documentation
- B. A mobile OS
- C. A relational database
- D. A CSS framework

Ans: A. Interactive OpenAPI/Swagger documentation

**Q246. Why is input validation important for a public API?**

[Medium]

- A. It has no security value
- B. It slows every request intentionally
- C. It rejects malformed or malicious data early, protecting the app and ensuring correct behavior
- D. It replaces the database

Ans: C. It rejects malformed or malicious data early, protecting the app and ensuring correct behavior

**Q247. A Streamlit prototype gets slow because expensive computation reruns on every widget interaction. What is the idiomatic fix?**

[Difficult]

- A. Delete all widgets
- B. Cache expensive results (e.g., with `st.cache_data` / `st.cache_resource`) so they aren't recomputed on each rerun
- C. Restart the server per click
- D. Move everything to a spreadsheet

Ans: B. Cache expensive results (e.g., with `st.cache_data` / `st.cache_resource`) so they aren't recomputed on each rerun

**Q248. A FastAPI service calling a slow external LLM blocks other requests under load. What is the correct approach?**

[Difficult]

- A. Make everything synchronous
- B. Add more print statements
- C. Use async endpoints and non-blocking I/O (`async/await`) so the server can handle concurrent requests while awaiting the LLM
- D. Disable concurrency entirely

Ans: C. Use async endpoints and non-blocking I/O (`async/await`) so the server can handle concurrent requests while awaiting the LLM

**Q249. Why should a public GenAI API implement rate limiting and authentication?**

[Difficult]

- A. Because APIs cannot be secured
- B. To prevent abuse/cost overrun, ensure fair usage, and control access to protected functionality and data
- C. To slow down all users equally for fun
- D. To remove documentation

Ans: B. To prevent abuse/cost overrun, ensure fair usage, and control access to protected functionality and data

**Q250. When serving an ML model behind an API, why is versioning the model and endpoint important?**

[Difficult]

- A. It enables safe rollouts, reproducibility, and rollback, and lets clients depend on stable behavior while you iterate
- B. Versioning slows the API
- C. Models never change
- D. It deletes old predictions

Ans: A. It enables safe rollouts, reproducibility, and rollback, and lets clients depend on stable behavior while you iterate

# Copilot Fundamentals

20 questions

**Q37. What is GitHub Copilot primarily designed to do?**

[Easy]

- A. Provide AI-powered code suggestions and completions inside the developer's editor
- B. Manage cloud billing invoices
- C. Replace the need for a version control system
- D. Serve as a relational database engine

Ans: A. Provide AI-powered code suggestions and completions inside the developer's editor

**Q38. In the context of AI coding assistants, what is a 'code suggestion'?**

[Easy]

- A. A ticket filed to the support team
- B. A compiled binary file
- C. AI-generated code the developer can accept, reject, or edit
- D. A network configuration setting

Ans: C. AI-generated code the developer can accept, reject, or edit

**Q39. Microsoft Copilot (M365 Copilot) is best described as:**

[Easy]

- A. An AI assistant embedded in productivity apps to help draft, summarize, and analyze content
- B. A hardware GPU accelerator
- C. A physical office badge system
- D. A database migration tool only

Ans: A. An AI assistant embedded in productivity apps to help draft, summarize, and analyze content

**Q40. How can GitHub Copilot assist with ETL automation for data engineering?**

[Medium]

- A. By encrypting all source data automatically
- B. By generating boilerplate transformation code (e.g., PySpark) and query snippets from natural-language comments, accelerating pipeline development
- C. By replacing the data warehouse entirely
- D. By physically installing servers

Ans: B. By generating boilerplate transformation code (e.g., PySpark) and query snippets from natural-language comments, accelerating pipeline development

**Q41. What is a realistic role for an AI assistant in generating SQL queries for a data engineer?**

[Medium]

- A. It permanently stores all query results in the model

- B. It drafts queries from a plain-language description that the engineer then validates and tests against the schema
- C. It guarantees every generated query is production-ready without review
- D. It executes queries directly on production without permissions

Ans: B. It drafts queries from a plain-language description that the engineer then validates and tests against the schema

**Q42. Why should developers review AI-generated code rather than accept it blindly?**

[Medium]

- A. Because Copilot cannot generate any correct code
- B. Suggestions may contain subtle bugs, insecure patterns, or logic that doesn't match business intent, so human validation and testing are essential
- C. AI code is always syntactically invalid
- D. Because reviewing is legally required to compile any program

Ans: B. Suggestions may contain subtle bugs, insecure patterns, or logic that doesn't match business intent, so human validation and testing are essential

**Q43. A team wants Copilot to help build a PySpark ETL job but is concerned about data governance and correctness. What is the most responsible workflow?**

[Difficult]

- A. Paste production credentials into prompts for convenience
- B. Use Copilot to accelerate drafting, then apply code review, unit/integration tests on sample data, schema validation, and governance checks before promoting to production
- C. Disable all testing to ship faster
- D. Let Copilot write and auto-deploy the pipeline to production untouched

Ans: B. Use Copilot to accelerate drafting, then apply code review, unit/integration tests on sample data, schema validation, and governance checks before promoting to production

**Q44. A data engineer uses an AI assistant to generate a transformation, but it silently drops rows with null keys, causing downstream data loss. What lesson does this illustrate about AI-assisted ETL?**

[Difficult]

- A. The assistant should be given write access to fix it live
- B. AI assistants understand full business context perfectly
- C. Null values are impossible in real data
- D. Generated code reflects only the prompt's stated intent; edge cases (nulls, duplicates, types) must be explicitly specified and verified, with data-quality checks in the pipeline

Ans: D. Generated code reflects only the prompt's stated intent; edge cases (nulls, duplicates, types) must be explicitly specified and verified, with data-quality checks in the pipeline

**Q140. How does GitHub Copilot typically receive context to generate a suggestion?**

[Easy]

- A. From the network cable length

- B. From the printer settings
- C. From the surrounding code and comments in the file the developer is editing
- D. From the monitor brightness

Ans: C. From the surrounding code and comments in the file the developer is editing

**Q141. Which of these is a valid way to prompt an AI coding assistant for a function?**

[Easy]

- A. Turning off the computer
- B. Writing a descriptive comment or function signature describing the desired behavior
- C. Deleting the project
- D. Renaming the operating system

Ans: B. Writing a descriptive comment or function signature describing the desired behavior

**Q142. What kind of tasks can Copilot-style assistants help with?**

[Easy]

- A. Managing office lighting
- B. Physically repairing hardware
- C. Autocompleting code, generating tests, and explaining snippets
- D. Brewing coffee

Ans: C. Autocompleting code, generating tests, and explaining snippets

**Q143. In Azure Data Factory, a Copilot-style assistant can help by:**

[Easy]

- A. Printing invoices
- B. Suggesting or generating data pipeline steps and transformations from natural-language intent
- C. Formatting the hard drive
- D. Replacing the electricity supply

Ans: B. Suggesting or generating data pipeline steps and transformations from natural-language intent

**Q144. What is a good practice when using Copilot for query generation against a specific schema?**

[Medium]

- A. Provide schema context and clear intent, then verify the generated query against the actual tables
- B. Run it on production immediately
- C. Assume the schema without checking
- D. Never test the query

Ans: A. Provide schema context and clear intent, then verify the generated query against the actual tables

**Q145. How can Copilot help a data engineer write PySpark transformations faster?**

[Medium]

- A. By deleting Spark
- B. By disabling the cluster
- C. By suggesting DataFrame transformation code from comments describing the desired output
- D. By converting Spark to a spreadsheet

Ans: C. By suggesting DataFrame transformation code from comments describing the desired output

**Q146. Why is context (open files, related code) important for Copilot's suggestion quality?**

[Medium]

- A. Context slows the model to zero
- B. Only the file name matters
- C. More relevant context helps the model produce suggestions aligned with the existing code and intent
- D. Context is ignored entirely

Ans: C. More relevant context helps the model produce suggestions aligned with the existing code and intent

**Q147. What is a limitation to keep in mind when Copilot generates code using external libraries?**

[Medium]

- A. It may suggest outdated APIs or nonexistent functions, so the code must be verified against current library docs
- B. It never uses libraries
- C. It always uses the newest APIs perfectly
- D. Generated library calls are guaranteed correct

Ans: A. It may suggest outdated APIs or nonexistent functions, so the code must be verified against current library docs

**Q148. How should sensitive secrets (API keys, passwords) be handled when using AI coding assistants?**

[Medium]

- A. Store them in comments
- B. Email them to the model
- C. Paste them directly for convenience
- D. Never paste secrets into prompts; keep them in environment variables or secret managers, not in code or prompts

Ans: D. Never paste secrets into prompts; keep them in environment variables or secret managers, not in code or prompts

**Q149. An organization worries Copilot-generated code may resemble licensed source. What governance step is prudent?**

[Difficult]

- A. Ignore licensing entirely

- B. Establish code-provenance/IP policies, use available duplication-detection/filtering settings, and require review to ensure license compliance
- C. Disable version control
- D. Publish all code publicly to avoid issues

Ans: B. Establish code-provenance/IP policies, use available duplication-detection/filtering settings, and require review to ensure license compliance

**Q150. A team adopts Copilot to speed ETL development but sees inconsistent code style and hidden bugs across the codebase. What is the best remedy?**

[Difficult]

- A. Let each developer accept all suggestions blindly
- B. Pair Copilot with linting, formatting standards, automated tests, and mandatory code review so AI accelerates work within guardrails
- C. Delete the tests
- D. Stop reviewing code to move faster

Ans: B. Pair Copilot with linting, formatting standards, automated tests, and mandatory code review so AI accelerates work within guardrails

**Q151. Why is it risky to let an AI assistant auto-generate and directly run migration scripts on a production database?**

[Difficult]

- A. Migrations never affect data
- B. Production databases cannot run scripts
- C. Generated scripts may contain destructive or incorrect operations; they require review, testing on non-prod, backups, and controlled deployment before touching production
- D. AI-generated scripts are always safe

Ans: C. Generated scripts may contain destructive or incorrect operations; they require review, testing on non-prod, backups, and controlled deployment before touching production

# Introduction to AI & ML

40 questions

## Q45. What is 'machine learning'?

[Easy]

- A. A method of manually coding every possible outcome
- B. A field where systems learn patterns from data to make predictions or decisions without being explicitly programmed for every rule
- C. A spreadsheet formula language
- D. A type of computer monitor

Ans: B. A field where systems learn patterns from data to make predictions or decisions without being explicitly programmed for every rule

## Q46. In supervised learning, the training data includes:

[Easy]

- A. Random noise only
- B. Only inputs with no labels
- C. No data at all
- D. Input features paired with known output labels

Ans: D. Input features paired with known output labels

## Q47. Which of the following is a typical unsupervised learning task?

[Easy]

- A. Forecasting demand from historical labeled sales
- B. Clustering customers into groups based on similarity without predefined labels
- C. Classifying emails using labeled spam/not-spam data
- D. Predicting house prices from labeled sales

Ans: B. Clustering customers into groups based on similarity without predefined labels

## Q48. What is 'deep learning' most closely associated with?

[Easy]

- A. Neural networks with many layers that learn hierarchical representations
- B. Manual rule writing
- C. Shallow decision trees only
- D. Relational database indexing

Ans: A. Neural networks with many layers that learn hierarchical representations

## Q49. What distinguishes classification from regression in supervised learning?

[Medium]

- A. Classification cannot use labeled data
- B. Classification predicts discrete categories; regression predicts continuous numeric values



- C. They are the same technique
- D. Regression only works on images

Ans: B. Classification predicts discrete categories; regression predicts continuous numeric values

**Q50. In reinforcement learning, how does an agent learn?**

[Medium]

- A. By interacting with an environment and receiving rewards or penalties to maximize cumulative reward
- B. By clustering data into groups
- C. By copying another model's weights directly
- D. By memorizing a fixed labeled dataset once

Ans: A. By interacting with an environment and receiving rewards or penalties to maximize cumulative reward

**Q51. What does 'overfitting' mean in machine learning?**

[Medium]

- A. The model trains too quickly
- B. The model is too simple to learn anything
- C. The dataset has too few columns
- D. The model performs well on training data but poorly on unseen data because it memorized noise instead of generalizable patterns

Ans: D. The model performs well on training data but poorly on unseen data because it memorized noise instead of generalizable patterns

**Q52. In NLP, what is 'tokenization'?**

[Medium]

- A. Breaking text into smaller units such as words or subwords for processing
- B. Compressing a file to save space
- C. Translating text into machine code
- D. Encrypting text with a secret key

Ans: A. Breaking text into smaller units such as words or subwords for processing

**Q53. A classifier reports 98% accuracy on a dataset where 98% of samples are the negative class. Why is accuracy misleading here and what is a better metric?**

[Difficult]

- A. Accuracy is always the best metric regardless of class balance
- B. Accuracy should be replaced by training time
- C. The dataset must be discarded entirely
- D. With severe class imbalance, a model predicting only the majority class scores high accuracy while failing the minority class; use precision, recall, F1, or AUC to evaluate meaningfully

Ans: D. With severe class imbalance, a model predicting only the majority class scores high accuracy while failing the minority class; use precision, recall, F1, or AUC to evaluate meaningfully

**Q54. A model shows low training error but high validation error. What is the diagnosis and a suitable remedy?**

[Difficult]

- A. Underfitting; add more layers only
- B. The learning rate is irrelevant to this
- C. The problem is always insufficient epochs
- D. Overfitting (high variance); remedies include regularization, more training data, simpler models, dropout, or early stopping

Ans: D. Overfitting (high variance); remedies include regularization, more training data, simpler models, dropout, or early stopping

**Q55. Why is splitting data into train/validation/test sets important, and what mistake causes 'data leakage'?**

[Difficult]

- A. The test set should be used to tune hyperparameters repeatedly
- B. Splitting is unnecessary if the model is deep enough
- C. Leakage means the disk ran out of space
- D. Splits give an honest estimate of generalization; leakage occurs when information from validation/test (or future data) influences training—e.g., fitting a scaler on the full dataset before splitting—inflating performance

Ans: D. Splits give an honest estimate of generalization; leakage occurs when information from validation/test (or future data) influences training—e.g., fitting a scaler on the full dataset before splitting—inflating performance

**Q56. In deep learning, what problem does the 'vanishing gradient' describe and one common mitigation?**

[Difficult]

- A. Gradients grow infinitely large, fixed by removing all layers
- B. It refers to losing the dataset file
- C. Gradients become extremely small in deep networks, slowing or stalling learning in early layers; mitigations include ReLU activations, residual connections, and normalization
- D. The model runs out of training labels

Ans: C. Gradients become extremely small in deep networks, slowing or stalling learning in early layers; mitigations include ReLU activations, residual connections, and normalization

**Q152. What is 'artificial intelligence' broadly?**

[Easy]

- A. A network cable standard
- B. The simulation of human-like intelligence by machines to perform tasks such as reasoning and perception
- C. A brand of computer monitor
- D. A type of spreadsheet

Ans: B. The simulation of human-like intelligence by machines to perform tasks such as reasoning and perception

**Q153. Which is an example of a supervised learning problem?**

[Easy]

- A. Randomly shuffling data
- B. Grouping unlabeled customers by similarity
- C. Predicting a house price from labeled historical sales
- D. Compressing an image

Ans: C. Predicting a house price from labeled historical sales

**Q154. What is a 'feature' in machine learning?**

[Easy]

- A. An input variable/attribute used by the model to make predictions
- B. The final prediction only
- C. A type of database
- D. A bug in the model

Ans: A. An input variable/attribute used by the model to make predictions

**Q155. What is a 'label' (target) in supervised learning?**

[Easy]

- A. A random number
- B. A column name only
- C. The learning rate
- D. The known correct output the model learns to predict

Ans: D. The known correct output the model learns to predict

**Q156. Which of these is a common clustering algorithm?**

[Easy]

- A. K-means
- B. Bubble sort
- C. Linear regression
- D. Binary search

Ans: A. K-means

**Q157. What is a neural network loosely inspired by?**

[Easy]

- A. Traffic lights
- B. Ocean tides
- C. Musical notes
- D. Interconnected neurons in the brain

Ans: D. Interconnected neurons in the brain

**Q158. What does 'training' a model mean?**

[Easy]

- A. Adjusting model parameters using data so it can make good predictions
- B. Deleting the dataset
- C. Buying a faster computer
- D. Printing the results

Ans: A. Adjusting model parameters using data so it can make good predictions

#### Q159. What is NLP (Natural Language Processing)?

[Easy]

- A. A database index
- B. A network protocol
- C. A field enabling computers to understand and generate human language
- D. A programming language

Ans: C. A field enabling computers to understand and generate human language

#### Q160. What is the purpose of a 'loss function'?

[Medium]

- A. To render charts
- B. To measure how far the model's predictions are from the true values so training can minimize it
- C. To connect to the internet
- D. To store the dataset

Ans: B. To measure how far the model's predictions are from the true values so training can minimize it

#### Q161. What is 'gradient descent'?

[Medium]

- A. An optimization algorithm that iteratively adjusts parameters in the direction that reduces the loss
- B. A type of database join
- C. A data storage format
- D. A clustering method

Ans: A. An optimization algorithm that iteratively adjusts parameters in the direction that reduces the loss

#### Q162. What is the difference between a parameter and a hyperparameter?

[Medium]

- A. Parameters are always integers
- B. Hyperparameters are learned automatically only
- C. Parameters are learned from data during training; hyperparameters are set before training (e.g., learning rate)
- D. They are the same

Ans: C. Parameters are learned from data during training; hyperparameters are set before training (e.g., learning rate)

**Q163. What is 'feature engineering'?**

[Medium]

- A. Encrypting the model
- B. Buying more hardware
- C. Deleting all features
- D. Creating or transforming input variables to improve model performance

Ans: D. Creating or transforming input variables to improve model performance

**Q164. In NLP, what does a word embedding capture?**

[Medium]

- A. The file size of the text
- B. The color of the text
- C. Semantic relationships between words as dense numeric vectors
- D. The font of the word

Ans: C. Semantic relationships between words as dense numeric vectors

**Q165. What is the role of an activation function in a neural network?**

[Medium]

- A. It stores the dataset
- B. It sorts the data
- C. It introduces non-linearity so the network can learn complex patterns
- D. It compresses images

Ans: C. It introduces non-linearity so the network can learn complex patterns

**Q166. What does 'cross-validation' help estimate?**

[Medium]

- A. The exact price of hardware
- B. How well a model generalizes, by training/testing on multiple data folds
- C. The color scheme
- D. The file encoding

Ans: B. How well a model generalizes, by training/testing on multiple data folds

**Q167. What distinguishes a convolutional neural network (CNN) as well-suited for images?**

[Medium]

- A. It only works on text
- B. Convolutional layers learn spatial/local patterns like edges and textures across the image
- C. It cannot process pixels
- D. It requires no training

Ans: B. Convolutional layers learn spatial/local patterns like edges and textures across the image

**Q168. What is 'transfer learning'?**

[Medium]

- A. Deleting the model
- B. Transferring money
- C. Reusing a model pre-trained on a large dataset and adapting it to a related task with less data
- D. Moving files between disks

Ans: C. Reusing a model pre-trained on a large dataset and adapting it to a related task with less data

**Q169. What does a confusion matrix summarize for a classifier?**

[Medium]

- A. True/false positives and negatives, enabling metrics like precision and recall
- B. The GPU temperature
- C. The training time only
- D. The dataset file size

Ans: A. True/false positives and negatives, enabling metrics like precision and recall

**Q170. What is the exploration vs exploitation trade-off in reinforcement learning?**

[Medium]

- A. Choosing a database engine
- B. Selecting a chart color
- C. Balancing trying new actions to learn (explore) against choosing known rewarding actions (exploit)
- D. Deciding file formats

Ans: C. Balancing trying new actions to learn (explore) against choosing known rewarding actions (exploit)

**Q171. A model has high bias and underfits. What is the correct interpretation and remedy?**

[Difficult]

- A. It memorized noise; add more regularization
- B. It is perfect and needs no change
- C. It is too simple to capture the pattern; increase model complexity, add features, or reduce regularization
- D. The data is leaking; remove the test set

Ans: C. It is too simple to capture the pattern; increase model complexity, add features, or reduce regularization

**Q172. Why is a validation set kept separate from the test set during model development?**

[Difficult]

- A. The test set is used to train
- B. The validation set tunes hyperparameters/model choices while the test set gives an unbiased final estimate on truly unseen data
- C. Validation data is discarded before training
- D. They serve identical purposes

Ans: B. The validation set tunes hyperparameters/model choices while the test set gives an unbiased final estimate on truly unseen data

**Q173. In an imbalanced fraud-detection problem, why might you optimize for recall on the positive class, accepting lower precision?**

[Difficult]

- A. The classes are perfectly balanced
- B. Missing a fraud (false negative) is far costlier than a false alarm, so catching more true frauds matters even with some false positives
- C. Precision is always irrelevant
- D. Recall reduces training time

Ans: B. Missing a fraud (false negative) is far costlier than a false alarm, so catching more true frauds matters even with some false positives

**Q174. What is regularization (e.g., L1/L2) intended to do?**

[Difficult]

- A. Speed up disk I/O
- B. Delete features randomly
- C. Penalize large/complex weights to reduce overfitting and improve generalization
- D. Increase overfitting deliberately

Ans: C. Penalize large/complex weights to reduce overfitting and improve generalization

**Q175. Why can a model that performs well offline degrade in production over time?**

[Difficult]

- A. Models never degrade
- B. Production has no data
- C. The GPU improves accuracy automatically
- D. Data/concept drift: real-world distributions change, so models need monitoring and periodic retraining

Ans: D. Data/concept drift: real-world distributions change, so models need monitoring and periodic retraining

**Q176. What is the vanishing role of the softmax function in a multi-class classifier's output layer?**

[Difficult]

- A. It sorts the classes alphabetically
- B. It converts raw scores (logits) into a probability distribution over classes summing to 1
- C. It reduces the dataset size
- D. It encrypts the output

Ans: B. It converts raw scores (logits) into a probability distribution over classes summing to 1

**Q177. A transformer model's 'attention' mechanism enables it to do what?**

[Difficult]

- A. Store data on disk
- B. Compress video
- C. Render user interfaces
- D. Weigh the relevance of different input tokens to each other, capturing context and long-range dependencies

Ans: D. Weigh the relevance of different input tokens to each other, capturing context and long-range dependencies

**Q178. Why is a held-out test set never used for feature selection or scaling decisions?**

[Difficult]

- A. Scaling must always use all data
- B. Using it leaks test information into modeling choices, producing optimistic, unreliable performance estimates
- C. Feature selection requires the test set
- D. It makes training faster

Ans: B. Using it leaks test information into modeling choices, producing optimistic, unreliable performance estimates

**Q179. What is the primary purpose of an epoch during neural network training?**

[Difficult]

- A. The final prediction
- B. A single data record
- C. The size of the model file
- D. One complete pass over the entire training dataset; multiple epochs let the model iteratively improve

Ans: D. One complete pass over the entire training dataset; multiple epochs let the model iteratively improve



# Generative AI Fundamentals

25 questions

## Q57. What is 'generative AI'?

[Easy]

- A. AI that only sorts existing files
- B. AI that can create new content such as text, images, audio, or code based on learned patterns
- C. A type of database index
- D. A hardware firewall

Ans: B. AI that can create new content such as text, images, audio, or code based on learned patterns

## Q58. What is a Large Language Model (LLM)?

[Easy]

- A. A spreadsheet macro
- B. A physical storage device
- C. A network routing protocol
- D. A model trained on vast text data that can understand and generate human-like language

Ans: D. A model trained on vast text data that can understand and generate human-like language

## Q59. In the context of GenAI, what is an 'AI agent'?

[Medium]

- A. A static text document
- B. A single fixed SQL query
- C. A system that can reason, plan, and use tools/actions to accomplish goals, often iterating over multiple steps
- D. A human customer-support representative

Ans: C. A system that can reason, plan, and use tools/actions to accomplish goals, often iterating over multiple steps

## Q60. What does 'hallucination' mean in the context of LLMs?

[Medium]

- A. The model translates languages
- B. The model refuses to answer
- C. The model runs out of memory
- D. The model generates plausible-sounding but incorrect or fabricated information

Ans: D. The model generates plausible-sounding but incorrect or fabricated information

## Q61. What is an 'embedding' in generative AI applications?

[Medium]

- A. A numerical vector representation of text or data that captures semantic meaning
- B. A way to compress video files

- C. A user authentication token
- D. A method for embedding logos into documents

Ans: A. A numerical vector representation of text or data that captures semantic meaning

**Q62. An enterprise wants an agentic workflow where an LLM autonomously calls internal tools/APIs to complete tasks. What is a critical design safeguard?**

[Difficult]

- A. Let the agent write directly to production databases without checks
- B. Remove all logging to improve speed
- C. Give the agent unrestricted access to all systems for flexibility
- D. Constrain the agent to governed, permissioned tools with validation, require confirmation for high-impact actions, and log every step for auditability and control

Ans: D. Constrain the agent to governed, permissioned tools with validation, require confirmation for high-impact actions, and log every step for auditability and control

**Q180. What kind of content can generative AI produce?**

[Easy]

- A. Text, images, audio, video, and code
- B. Only network packets
- C. Only printed paper
- D. Only spreadsheets

Ans: A. Text, images, audio, video, and code

**Q181. What is a 'token' in a language model?**

[Easy]

- A. A unit of text (word or word fragment) the model processes
- B. A database row
- C. A network port
- D. A physical security key

Ans: A. A unit of text (word or word fragment) the model processes

**Q182. What is a 'foundation model'?**

[Easy]

- A. A concrete building base
- B. A spreadsheet template
- C. A large model pre-trained on broad data that can be adapted to many tasks
- D. A single SQL table

Ans: C. A large model pre-trained on broad data that can be adapted to many tasks

**Q183. What is 'fine-tuning' a model?**

[Easy]

- A. Deleting the model

- B. Further training a pre-trained model on specific data to specialize it
- C. Compressing images
- D. Renaming files

Ans: B. Further training a pre-trained model on specific data to specialize it

**Q184. What is a 'context window' in an LLM?**

[Easy]

- A. A database index
- B. The maximum amount of text (tokens) the model can consider at once
- C. The GPU fan speed
- D. The screen brightness

Ans: B. The maximum amount of text (tokens) the model can consider at once

**Q185. What is 'multimodal' AI?**

[Easy]

- A. A backup system
- B. A model that can process and/or generate more than one type of data (e.g., text and images)
- C. A model with one input only
- D. A type of monitor

Ans: B. A model that can process and/or generate more than one type of data (e.g., text and images)

**Q186. What does the 'temperature' parameter control in text generation?**

[Medium]

- A. The randomness/creativity of outputs—lower is more deterministic, higher is more varied
- B. The document font
- C. The GPU heat
- D. The file size

Ans: A. The randomness/creativity of outputs—lower is more deterministic, higher is more varied

**Q187. Why do LLMs sometimes produce confident but false statements?**

[Medium]

- A. They intentionally lie
- B. They only output numbers
- C. They have no training data
- D. They predict likely text from patterns and lack grounded truth verification, leading to hallucinations

Ans: D. They predict likely text from patterns and lack grounded truth verification, leading to hallucinations

**Q188. What is the transformer architecture known for in generative AI?**

[Medium]

- A. Managing databases

- B. Rendering 3D games
- C. Using self-attention to model relationships across sequences, powering modern LLMs
- D. Storing files

Ans: C. Using self-attention to model relationships across sequences, powering modern LLMs

**Q189. What is 'pre-training' versus 'fine-tuning'?**

[Medium]

- A. Pre-training uses no data
- B. Fine-tuning comes before pre-training always
- C. They are identical
- D. Pre-training learns broad patterns from large general data; fine-tuning adapts the model to a narrower task/domain

Ans: D. Pre-training learns broad patterns from large general data; fine-tuning adapts the model to a narrower task/domain

**Q190. What is a practical reason enterprises use smaller models for some tasks?**

[Medium]

- A. Model size never affects cost
- B. Lower cost and latency when the task is simple enough, reserving large models for complex needs
- C. Large models cannot run at all
- D. Small models are always more accurate

Ans: B. Lower cost and latency when the task is simple enough, reserving large models for complex needs

**Q191. What is RLHF (Reinforcement Learning from Human Feedback) used for?**

[Medium]

- A. Aligning model behavior with human preferences by training on human-rated responses
- B. Compressing the model
- C. Encrypting prompts
- D. Building databases

Ans: A. Aligning model behavior with human preferences by training on human-rated responses

**Q192. What is 'grounding' an LLM response?**

[Medium]

- A. Turning off the model
- B. Basing the answer on authoritative provided data/context to improve accuracy and reduce hallucination
- C. Randomizing the output
- D. Deleting the context

Ans: B. Basing the answer on authoritative provided data/context to improve accuracy and reduce hallucination

**Q193. What is an 'AI orchestration' layer in a GenAI application?**

[Medium]

- A. A managed flow coordinating prompts, retrieval, tools, filtering, and model calls into one pipeline
- B. A backup disk
- C. A single hardware chip
- D. A spreadsheet macro

Ans: A. A managed flow coordinating prompts, retrieval, tools, filtering, and model calls into one pipeline

**Q194. An agent loops indefinitely calling tools without converging on an answer. What design controls address this?**

[Difficult]

- A. Remove all tools permanently
- B. Disable logging
- C. Set iteration/step limits, add stopping criteria, monitor tool-call budgets, and include reflection or a supervisor to detect non-progress
- D. Give it infinite time

Ans: C. Set iteration/step limits, add stopping criteria, monitor tool-call budgets, and include reflection or a supervisor to detect non-progress

**Q195. Why is evaluating GenAI output harder than a traditional classification metric?**

[Difficult]

- A. GenAI cannot be evaluated
- B. Accuracy alone fully captures quality
- C. Outputs are open-ended and can be correct in many forms, so evaluation needs task-specific rubrics, human review, or reference-based/LLM-as-judge methods
- D. There is only one correct output always

Ans: C. Outputs are open-ended and can be correct in many forms, so evaluation needs task-specific rubrics, human review, or reference-based/LLM-as-judge methods

**Q196. What is a key risk of sending proprietary enterprise data to an external foundation model, and a mitigation?**

[Difficult]

- A. There is no risk at all
- B. Encryption is impossible
- C. The model always memorizes and republishes data
- D. Data exposure/leakage; mitigate with contractual no-training terms, data masking, private deployment, and minimizing sensitive data in prompts

Ans: D. Data exposure/leakage; mitigate with contractual no-training terms, data masking, private deployment, and minimizing sensitive data in prompts

**Q197. When should an enterprise prefer fine-tuning over prompting/RAG?**

[Difficult]

- A. Never, prompting always suffices

- B. Only to reduce accuracy
- C. Whenever knowledge changes hourly
- D. When it needs consistent specialized behavior/format or domain style that prompting can't reliably achieve, and the knowledge is relatively stable

Ans: D. When it needs consistent specialized behavior/format or domain style that prompting can't reliably achieve, and the knowledge is relatively stable

**Q198. Why might combining RAG with a fine-tuned model outperform either alone for enterprise support?**

[Difficult]

- A. Fine-tuning instills style/behavior while RAG supplies up-to-date, grounded facts—together improving both consistency and freshness
- B. They conflict and cannot be combined
- C. Fine-tuning provides real-time facts by itself
- D. RAG removes the need for any model

Ans: A. Fine-tuning instills style/behavior while RAG supplies up-to-date, grounded facts—together improving both consistency and freshness

# Prompt Engineering

20 questions

## Q63. What is a 'prompt' in generative AI?

[Easy]

- A. The input text or instruction given to an AI model to guide its response
- B. A model configuration file
- C. An error message from the server
- D. A scheduled system reminder

Ans: A. The input text or instruction given to an AI model to guide its response

## Q64. What is 'few-shot prompting'?

[Medium]

- A. Sending an empty prompt
- B. Providing a few examples of the desired input-output behavior in the prompt to guide the model
- C. Disabling the model's reasoning
- D. Fine-tuning the model on millions of records

Ans: B. Providing a few examples of the desired input-output behavior in the prompt to guide the model

## Q65. Why does giving an LLM a clear role, explicit constraints, and desired output format usually improve results?

[Medium]

- A. It only affects response speed, not quality
- B. It reduces the model's vocabulary permanently
- C. It guarantees the model will never make mistakes
- D. Specific instructions reduce ambiguity, steering the model toward the intended task, tone, and structure

Ans: D. Specific instructions reduce ambiguity, steering the model toward the intended task, tone, and structure

## Q66. A reasoning-heavy task yields wrong final answers, but asking the model to 'think step by step' improves accuracy. What technique is this and why does it help?

[Difficult]

- A. Data masking; it hides the question
- B. Chain-of-thought prompting; eliciting intermediate reasoning steps helps the model decompose the problem before committing to a final answer
- C. Zero-shot classification; it compresses the prompt
- D. Temperature scaling; it removes reasoning

Ans: B. Chain-of-thought prompting; eliciting intermediate reasoning steps helps the model decompose the problem before committing to a final answer

**Q67. A production prompt works in testing but breaks when users paste arbitrary text that overrides its instructions. What is this risk and a sound mitigation?**

[Difficult]

- A. Prompt injection; mitigate by separating trusted instructions from untrusted user input, validating/sanitizing inputs, and enforcing access controls outside the prompt rather than relying on instructions alone
- B. It is a font-rendering bug; change the CSS
- C. It is a hardware fault; add more RAM
- D. It is impossible; LLMs ignore user text

Ans: A. Prompt injection; mitigate by separating trusted instructions from untrusted user input, validating/sanitizing inputs, and enforcing access controls outside the prompt rather than relying on instructions alone

**Q199. What is 'zero-shot prompting'?**

[Easy]

- A. Sending 100 examples
- B. Deleting the prompt
- C. Asking the model to do a task with no examples, relying on its pre-trained knowledge
- D. Turning the model off

Ans: C. Asking the model to do a task with no examples, relying on its pre-trained knowledge

**Q200. Which is a good habit when writing prompts?**

[Easy]

- A. Be as vague as possible
- B. Be clear and specific about the task, audience, and desired format
- C. Never state the goal
- D. Use only emojis

Ans: B. Be clear and specific about the task, audience, and desired format

**Q201. What is a 'system prompt' typically used for?**

[Easy]

- A. Sending email
- B. Setting the model's role, rules, and behavior for the conversation
- C. Formatting a hard drive
- D. Rebooting the server

Ans: B. Setting the model's role, rules, and behavior for the conversation

**Q202. Why include the desired output format (e.g., JSON, bullet list) in a prompt?**

[Easy]

- A. It slows the model
- B. It guides the model to return results in a structure that's easy to use downstream
- C. It deletes the response



D. It has no effect

Ans: B. It guides the model to return results in a structure that's easy to use downstream

**Q203. What is the benefit of providing examples (few-shot) in a prompt?**

[Medium]

- A. It demonstrates the desired pattern, improving consistency and accuracy on the task
- B. It always retrains the model
- C. It disables reasoning
- D. It reduces the context window to zero

Ans: A. It demonstrates the desired pattern, improving consistency and accuracy on the task

**Q204. How does asking the model to 'cite the source or say I don't know' help?**

[Medium]

- A. It guarantees creativity
- B. It speeds up the GPU
- C. It reduces hallucination by encouraging grounded, verifiable answers and honest abstention
- D. It removes all context

Ans: C. It reduces hallucination by encouraging grounded, verifiable answers and honest abstention

**Q205. What is 'prompt chaining'?**

[Medium]

- A. Encrypting prompts
- B. Deleting intermediate steps
- C. Sending one giant prompt only
- D. Breaking a complex task into a sequence of prompts where each step's output feeds the next

Ans: D. Breaking a complex task into a sequence of prompts where each step's output feeds the next

**Q206. Why is it useful to specify constraints (length, tone, audience) in a prompt?**

[Medium]

- A. They disable the model
- B. Constraints confuse the model
- C. They only affect color
- D. Constraints narrow the output space so results match requirements more reliably

Ans: D. Constraints narrow the output space so results match requirements more reliably

**Q207. What does providing 'delimiters' (e.g., triple quotes) around user content achieve?**

[Medium]

- A. It encrypts the content
- B. It shortens the model
- C. It deletes the input
- D. It clearly separates instructions from data, reducing ambiguity and some injection risk

Ans: D. It clearly separates instructions from data, reducing ambiguity and some injection risk

**Q208. What is a reasonable use of a low temperature in prompting?**

[Medium]

- A. Disabling the model
- B. Creative brainstorming only
- C. Tasks needing consistent, factual, deterministic answers such as data extraction
- D. Random poetry

Ans: C. Tasks needing consistent, factual, deterministic answers such as data extraction

**Q209. A complex analytical prompt gives shallow answers. Which combination of techniques most improves depth and reliability?**

[Difficult]

- A. Decompose with prompt chaining, add few-shot exemplars, request step-by-step reasoning, and specify a structured output with self-check
- B. Send an empty prompt
- C. Ask only yes/no questions
- D. Increase temperature and remove all instructions

Ans: A. Decompose with prompt chaining, add few-shot exemplars, request step-by-step reasoning, and specify a structured output with self-check

**Q210. Why can chain-of-thought reasoning sometimes be hidden or summarized rather than shown to end users in production?**

[Difficult]

- A. Because users cannot read text
- B. Because it deletes the answer
- C. Because reasoning is always wrong
- D. To avoid exposing intermediate reasoning that may be verbose, sensitive, or exploitable, while still benefiting from the reasoning internally

Ans: D. To avoid exposing intermediate reasoning that may be verbose, sensitive, or exploitable, while still benefiting from the reasoning internally

**Q211. A prompt reliably works on one model but fails on another. What is the sound engineering response?**

[Difficult]

- A. Delete the failing model's users
- B. Never change the prompt
- C. Recognize prompts are model-sensitive; re-test and adapt prompts per model, and maintain evaluation sets to catch regressions
- D. Assume all models behave identically

Ans: C. Recognize prompts are model-sensitive; re-test and adapt prompts per model, and maintain evaluation sets to catch regressions

**Q212. How does 'self-consistency' improve answers on reasoning tasks?**

[Difficult]

- A. It sends one deterministic answer only
- B. It removes reasoning
- C. Sampling multiple reasoning paths and taking the majority/most-consistent answer reduces the impact of any single flawed chain
- D. It shortens the prompt

Ans: C. Sampling multiple reasoning paths and taking the majority/most-consistent answer reduces the impact of any single flawed chain

**Q213. Why should you never rely solely on prompt instructions like 'do not reveal secrets' for security?**

[Difficult]

- A. Because the model always obeys perfectly
- B. Because security is unnecessary
- C. Instructions can be overridden by injection or clever inputs; enforce security at the system/data/access layer independent of the prompt
- D. Because prompts are encrypted

Ans: C. Instructions can be overridden by injection or clever inputs; enforce security at the system/data/access layer independent of the prompt

# GenAI Tools

15 questions

**Q68. Which of the following is a well-known text-based generative AI assistant?**

[Easy]

- A. WinRAR
- B. MySQL Workbench
- C. Adobe Reader
- D. ChatGPT

Ans: D. ChatGPT

**Q69. DALL·E is primarily known for which capability?**

[Easy]

- A. Compiling C++ code
- B. Running SQL queries
- C. Generating images from text prompts
- D. Managing cloud storage buckets

Ans: C. Generating images from text prompts

**Q70. Anthropic's Claude and Mistral's models are examples of what?**

[Medium]

- A. Foundation/large language models offered by different AI providers
- B. Relational databases
- C. Spreadsheet applications
- D. Operating systems

Ans: A. Foundation/large language models offered by different AI providers

**Q71. When selecting among GenAI tools/models for an enterprise workload, which combination of factors is most appropriate to weigh?**

[Difficult]

- A. Solely the largest model regardless of cost or privacy
- B. Task fit and quality, context window, cost/latency, data privacy and governance terms, and integration/deployment options
- C. The color of the tool's logo
- D. Only the model's brand name

Ans: B. Task fit and quality, context window, cost/latency, data privacy and governance terms, and integration/deployment options

**Q214. Which tool is primarily a conversational text assistant?**

[Easy]

- A. Notepad

- B. Excel
- C. Photoshop
- D. ChatGPT

Ans: D. ChatGPT

**Q215. Which capability is Midjourney/DALL·E associated with?**

[Easy]

- A. Database backups
- B. Text-to-image generation
- C. Spreadsheet formulas
- D. Network routing

Ans: B. Text-to-image generation

**Q216. Claude, made by Anthropic, is best described as:**

[Easy]

- A. An operating system
- B. A large language model / AI assistant
- C. A relational database
- D. A web browser

Ans: B. A large language model / AI assistant

**Q217. What is a common way developers access GenAI models in applications?**

[Easy]

- A. By printing prompts
- B. Through an API provided by the model vendor
- C. By mailing a hard drive
- D. Through a fax machine

Ans: B. Through an API provided by the model vendor

**Q218. What is an advantage of using an API to access a hosted LLM versus running your own?**

[Medium]

- A. It requires buying GPUs
- B. It guarantees zero latency
- C. It removes all costs
- D. No infrastructure to manage, easy scaling, and access to updated models—at the cost of ongoing usage fees and data-handling considerations

Ans: D. No infrastructure to manage, easy scaling, and access to updated models—at the cost of ongoing usage fees and data-handling considerations

**Q219. When comparing GenAI tools, why does context window size matter?**

[Medium]

- A. A larger context window lets the model consider more input (long documents, more history) in a single request
- B. It sets the price to zero
- C. It only affects images
- D. It changes the logo color

Ans: A. A larger context window lets the model consider more input (long documents, more history) in a single request

**Q220. Why might an enterprise prefer an open-weights model deployed privately?**

[Medium]

- A. Open models cannot be customized
- B. Greater control over data residency, customization, and no external data sharing—at the cost of managing infrastructure
- C. It removes all maintenance
- D. It always guarantees higher accuracy

Ans: B. Greater control over data residency, customization, and no external data sharing—at the cost of managing infrastructure

**Q221. What is a 'playground' or console offered by GenAI providers used for?**

[Medium]

- A. Managing payroll
- B. Experimenting with prompts and parameters interactively before integrating via API
- C. Editing photos only
- D. Playing video games

Ans: B. Experimenting with prompts and parameters interactively before integrating via API

**Q222. A team must choose between a general chat model and a code-specialized tool for a dev workflow. What is the most rational basis?**

[Difficult]

- A. Never evaluate; choose randomly
- B. Match the tool to the task—code assistants integrate with editors and excel at coding, while general models suit broad reasoning—also weighing cost, privacy, and integration
- C. Pick by logo color
- D. Always pick whichever has more marketing

Ans: B. Match the tool to the task—code assistants integrate with editors and excel at coding, while general models suit broad reasoning—also weighing cost, privacy, and integration

**Q223. Why is vendor lock-in a consideration when standardizing on one GenAI provider?**

[Difficult]

- A. APIs never change
- B. Lock-in has no downside
- C. Heavy coupling to one provider's APIs/features can raise switching costs; abstraction layers and evaluations across providers preserve flexibility

D. All providers are identical

Ans: C. Heavy coupling to one provider's APIs/features can raise switching costs; abstraction layers and evaluations across providers preserve flexibility

**Q224. When integrating a third-party GenAI API into a regulated product, which factors are essential to review?**

**[Difficult]**

A. Only the response speed

B. Only the logo

C. Data-handling and retention terms, compliance certifications, region/residency, uptime SLAs, and content/safety controls

D. Only the marketing claims

Ans: C. Data-handling and retention terms, compliance certifications, region/residency, uptime SLAs, and content/safety controls

# Retrieval Augmented Generation (RAG)

22 questions

## Q72. What does RAG (Retrieval Augmented Generation) do?

[Easy]

- A. Retrieves relevant external/enterprise information and supplies it to the LLM as context to generate grounded answers
- B. Only works with images
- C. Retrains the LLM from scratch on every query
- D. Deletes the knowledge base before answering

Ans: A. Retrieves relevant external/enterprise information and supplies it to the LLM as context to generate grounded answers

## Q73. What is the role of a vector store in a RAG pipeline?

[Medium]

- A. To compile the application code
- B. To render the user interface
- C. To store user passwords
- D. To store embeddings and enable fast similarity search to retrieve semantically relevant chunks

Ans: D. To store embeddings and enable fast similarity search to retrieve semantically relevant chunks

## Q74. Why is RAG often preferred over fine-tuning for frequently changing knowledge?

[Medium]

- A. RAG bakes the knowledge permanently into the weights
- B. Fine-tuning updates instantly whenever a document changes
- C. RAG requires no data at all
- D. RAG lets you update the knowledge base without retraining the model, suiting information that changes often

Ans: D. RAG lets you update the knowledge base without retraining the model, suiting information that changes often

## Q75. What does 'semantic similarity' measure when retrieving chunks in RAG?

[Medium]

- A. The order documents were uploaded
- B. The exact number of matching characters
- C. How close two pieces of content are in meaning, computed as distance between their vector embeddings
- D. The file size difference between documents

Ans: C. How close two pieces of content are in meaning, computed as distance between their vector embeddings



**Q76. A RAG assistant retrieves the right documents but its answers subtly contradict the sources. What layered mitigation improves faithfulness?**

[Difficult]

- A. Remove retrieval and rely on the base model
- B. Shorten every document to one sentence
- C. Instruct the model to answer only from provided context, add a grounding/faithfulness check against retrieved passages, require inline citations, and suppress answers not traceable to evidence
- D. Raise temperature so the model is more confident

Ans: C. Instruct the model to answer only from provided context, add a grounding/faithfulness check against retrieved passages, require inline citations, and suppress answers not traceable to evidence

**Q77. In a RAG system, retrieval quality is poor because chunks are either too large (diluted) or too small (fragmented). What is the sound engineering approach?**

[Difficult]

- A. Tune chunk size (and overlap) to the document type and query patterns—balancing enough context per chunk for meaning against small enough chunks for precise retrieval—and evaluate with a representative test set
- B. Always use the entire document as a single chunk
- C. Always use one-word chunks
- D. Chunk size never affects retrieval quality

Ans: A. Tune chunk size (and overlap) to the document type and query patterns—balancing enough context per chunk for meaning against small enough chunks for precise retrieval—and evaluate with a representative test set

**Q225. In RAG, what is 'retrieval'?**

[Easy]

- A. Compiling code
- B. Fetching relevant documents/chunks from a knowledge source to inform the model's answer
- C. Deleting the model
- D. Generating random text

Ans: B. Fetching relevant documents/chunks from a knowledge source to inform the model's answer

**Q226. What is 'chunking' in a RAG pipeline?**

[Easy]

- A. Merging all files into one token
- B. Encrypting the database
- C. Deleting documents
- D. Splitting documents into smaller passages for embedding and retrieval

Ans: D. Splitting documents into smaller passages for embedding and retrieval

**Q227. What is stored in a vector database for RAG?**

[Easy]

- A. Embeddings (numeric vectors) of content, enabling similarity search
- B. Compiled binaries
- C. Only images
- D. User passwords only

Ans: A. Embeddings (numeric vectors) of content, enabling similarity search

### Q228. Why do RAG systems reduce hallucination compared to a plain LLM?

[Easy]

- A. They never use an LLM
- B. They ground answers in retrieved, relevant source content rather than the model's memory alone
- C. They disable the model
- D. They only answer yes/no

Ans: B. They ground answers in retrieved, relevant source content rather than the model's memory alone

### Q229. What is an embedding model's job in RAG?

[Medium]

- A. To render the UI
- B. To convert text into vectors that place semantically similar content close together
- C. To store passwords
- D. To compile queries

Ans: B. To convert text into vectors that place semantically similar content close together

### Q230. What distance/similarity measure is commonly used to compare embeddings?

[Medium]

- A. Cosine similarity (or dot product / Euclidean distance)
- B. File modification date
- C. Alphabetical order
- D. RGB color distance

Ans: A. Cosine similarity (or dot product / Euclidean distance)

### Q231. What is 'top-k' retrieval in RAG?

[Medium]

- A. Deleting k chunks
- B. Returning every chunk always
- C. Sorting by file size
- D. Returning the k most relevant chunks to include as context

Ans: D. Returning the k most relevant chunks to include as context

### Q232. Why is chunk overlap sometimes used?

[Medium]

- A. To waste storage deliberately
- B. To preserve context that spans chunk boundaries, avoiding cutting off relevant information
- C. To slow retrieval
- D. To duplicate the whole database

Ans: B. To preserve context that spans chunk boundaries, avoiding cutting off relevant information

**Q233. What is a 're-ranker' in an advanced RAG pipeline?**

[Medium]

- A. A second-stage model that reorders retrieved candidates by relevance for better final context
- B. A password manager
- C. A UI theme
- D. A backup tool

Ans: A. A second-stage model that reorders retrieved candidates by relevance for better final context

**Q234. Why might hybrid search (keyword + vector) outperform vector search alone?**

[Medium]

- A. It combines exact keyword matching with semantic similarity, improving recall for both precise terms and conceptual queries
- B. Vector search cannot be combined with anything
- C. Hybrid search ignores relevance
- D. Keyword search is always better

Ans: A. It combines exact keyword matching with semantic similarity, improving recall for both precise terms and conceptual queries

**Q235. What is metadata filtering in retrieval used for?**

[Medium]

- A. Deleting metadata
- B. Restricting retrieval to chunks matching attributes (e.g., department, date, access level) before/along with similarity search
- C. Rendering charts
- D. Encrypting all data

Ans: B. Restricting retrieval to chunks matching attributes (e.g., department, date, access level) before/along with similarity search

**Q236. A RAG assistant returns outdated answers even after documents were updated. What is the likely cause and fix?**

[Difficult]

- A. RAG cannot update ever
- B. The LLM is broken forever
- C. The vector index wasn't refreshed; implement a re-embedding/ingestion pipeline that updates the index when source documents change
- D. Delete the knowledge base permanently

Ans: C. The vector index wasn't refreshed; implement a re-embedding/ingestion pipeline that updates the index when source documents change

**Q237. Retrieval returns semantically similar but wrong-scope chunks (e.g., another product line). Which fix is most targeted?**

[Difficult]

- A. Merge all products into one document
- B. Add metadata filtering and namespaces so retrieval is constrained to the correct scope, plus improve chunking/embeddings
- C. Remove retrieval
- D. Increase temperature

Ans: B. Add metadata filtering and namespaces so retrieval is constrained to the correct scope, plus improve chunking/embeddings

**Q238. How should access control be enforced in an enterprise RAG system so users only see permitted content?**

[Difficult]

- A. Apply permission filters at retrieval time (and at the data layer), never relying on the prompt to hide unauthorized data
- B. Trust the LLM to hide restricted data
- C. Give all users all documents
- D. Disable authentication

Ans: A. Apply permission filters at retrieval time (and at the data layer), never relying on the prompt to hide unauthorized data

**Q239. What does measuring 'retrieval precision and recall' tell you about a RAG system?**

[Difficult]

- A. Whether retrieved chunks are relevant (precision) and whether relevant chunks are being found (recall)—key to diagnosing answer quality
- B. The GPU temperature
- C. The UI color
- D. The user's password strength

Ans: A. Whether retrieved chunks are relevant (precision) and whether relevant chunks are being found (recall)—key to diagnosing answer quality

**Q240. Why can very large top-k retrieval hurt answer quality and cost?**

[Difficult]

- A. It reduces cost to zero
- B. Too much context can dilute relevant signal, increase token cost/latency, and cause the model to lose focus; tune k and use re-ranking
- C. More context is always strictly better
- D. It has no effect at all

Ans: B. Too much context can dilute relevant signal, increase token cost/latency, and cause the model to lose focus; tune k and use re-ranking